



**Assignment 1.2: AI Systems & Security Requirements Report**

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## 1. Engagement Overview

Design World has initiated an AI security engagement to support the planned deployment of an enterprise AI platform. The platform is intended to improve employee productivity and support approved business and engineering workflows. This engagement focuses on defining the system and security requirements that must be understood before the organization moves into architecture design.

The scope of this report is to define the security requirements for the AI platform before it is deployed into production and is ready for use operationally. It identifies what the AI platform is expected to do and what information and capabilities it must protect while also supporting business workflows. This report does not select technical controls, propose architecture designs, or perform threat modeling.

This engagement is important because Design World does not only need an AI tool that functions properly; it needs an AI platform that can be used while also maintaining governance of the platform. Due to the intended use cases for the platform, clear requirements are necessary to make sure the system's purpose and operating constraints are understood before the platform is deployed into production.

The requirements established in this report provide the baseline for the remaining AI security engagement project.

## 2. Business Objectives and Mission Goals

Design World intends to deploy an enterprise AI platform that improves employee productivity, organizational knowledge access, and business workflow efficiency. The platform should support approved business activities without replacing established decision-making authority and accountability/corporate oversight.

The following business and mission objectives define outcomes the AI platform is expected to support:

### **BMO-1: Improve Access to Organizational Knowledge**

The AI platform should provide employees with a more efficient method of locating, understanding, and using information from approved enterprise data sources. Users should be able to ask natural-language questions and receive relevant responses without manually searching across multiple systems/repositories.

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### **BMO-2: Increase Workforce Productivity**

The platform should reduce the time employees spend performing repetitive knowledge-based tasks. This includes searching for information, summarizing documents, preparing initial drafts, and organizing business or technical content. The platform should allow employees to focus more attention on analysis, collaboration, and higher-value responsibilities.

### **BMO-3: Support Consistent Business and Engineering Work Products**

The AI platform should assist employees in producing clear and consistent work products based on approved organizational information. Associated work products may include summaries, reports, technical documentation, and project materials.

AI-generated work products should support employees rather than replace professional judgment, required reviews, or existing approval processes.

### **BMO-4: Improve Approved Business Workflows**

Design World intends to use the platform to reduce operational friction in approved workflows. The AI platform may assist with routine tasks and initiate authorized actions measurable business value is applicable.

Any workflow assistance must remain consistent with the purpose of the approved business process. Overall Workflow should not bypass required human review, business ownership, or organizational accountability.

### **BMO-6: Enable Responsible Enterprise AI Adoption**

The platform should provide Design World with a governed alternative to unvetted AI tools. The organization's objective is to obtain the productivity benefits of AI while maintaining confidence that the platform is being used for authorized purposes, within the organization's accepted level of risk.

### **BMO-7: Produce Measurable Business Value**

The platform should demonstrate measurable improvements in areas such as information retrieval time, task completion time, quality of work products, employee adoption, and user satisfaction.

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Business value should be evaluated against defined success criteria. This allows leadership to determine whether the platform continues to support Design World's mission + operational priorities.

### 3. AI Use Case and Functional Expectations

#### a. AI Use Cases

##### UC-1: Enterprise Knowledge Retrieval

Authorized users should be able to ask questions about approved internal information to receive relevant/understandable responses. This use case is intended to reduce the time required to locate information across organizational documents, repositories, policies, and other enterprise data sources.

##### UC-2: Document Summarization and Information

The platform should be able to summarize approved documents and combine relevant information from multiple authorized sources. Users may use this capability to identify key points, compare information, or understand large amounts of organizational content.

##### UC-3: Business and Technical Content Assistance

The platform should assist users in drafting, revising, and organizing internal business materials. Examples: project documentation, meeting summaries, procedural content, and preliminary engineering work products.

All generated content should be treated as an assistive work product that should have the appropriate employee review and approval.

##### UC-4: Controlled Workflow Assistance

The platform may support approved workflows by preparing information, initiating routine requests, or performing specific authorized actions. These activities should remain limited to defined business purposes and should not be treated as unrestricted decision-making.

#### b. AI Functional Expectations

##### FE-1: Natural-Language Interaction

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Users should be able to interact with the platform with clear conversational prompts, without needing specialized technical knowledge. The platform should interpret user requests and provide responses appropriately for the requested business task.

#### **FE-2: Relevant and Context-Aware Responses**

The platform should provide responses that are relevant to the user's request and appropriate to the available organizational context. Responses should distinguish between information supported by approved enterprise sources and any external sources.

#### **FE-3: Support for Approved Enterprise Information**

The platform should retrieve and only process applicable information approved by the organization. The availability of information should remain consistent with the organization's permissions.

#### **FE-4: Human Review and Decision Authority**

Users must remain responsible for reviewing AI-generated work products and approving decisions. Users are also responsible for actions that require professional or organizational authority.

#### **FE-5: Measurable Use-Case Effectiveness**

Design World should be able to evaluate whether each approved use case is achieving the intended business outcome. Evaluation criteria may include the usefulness of the response, task completion time, work product quality, frequency of required corrections, and user satisfaction.

### **4. Security Objectives (confidentiality, integrity, availability, and safety)**

Design World must define the security outcomes the AI platform is expected to support. These objectives establish what the platform must protect and how it should behave when handling internal data and generating responses.

Please refer below for the AI platforms security objectives:

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### **SO-1: Data Confidentiality and Privacy Protection**

The AI platform must enforce strict access controls to prevent the unauthorized disclosure or exposure of Design World's intellectual property.

### **SO-2: Output Integrity and Trustworthiness**

The platform must generate reliable, accurate work products that users can confidently depend on for business and engineering decisions. System responses must align strictly with approved enterprise sources, and the architecture must include safeguards to mitigate the risks of misinformation.

### **SO-3: Operational Resilience and System Availability**

The AI infrastructure must be reliable enough to support day-to-day business operations without introducing downtime or performance bottlenecks. Computational workloads driven by the platform must not degrade the availability, responsiveness, or stability of the system it is integrated with.

### **SO-4: Constrained Execution and Authorization Boundaries**

Any autonomous actions or API calls executed by the AI platform on a user's behalf must stay within the intended scope and the user's explicit permissions. The platform must be designed to prevent privilege escalation, unauthorized system modifications, or unintended business transactions that bypass standard corporate oversight.

### **SO-5: Governance, Auditability, and Traceability**

Design World must maintain full visibility into how the AI platform is used and how specific outcomes are reached. The system must support comprehensive logging and explainable decision paths, enabling the organization to audit all activity from the AI platform.

### **SO-6: Alignment with Data Governance**

The platform must handle information according to Design World's established data classification and handling frameworks. Internal data should not be retrieved, stored, or shared in a way that bypasses any data classification rules that would normally apply to that information.

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## 5. Governance and Policy Requirements

Design World requires an AI-specific governance approach before the platform can be launched. The case study identifies a governance gap, meaning the organization does not currently have a complete AI-specific acceptance process. This makes governance a central requirement for the engagement.

The platform must operate under approved business use cases and within the organization's risk level. Governance requirements should clarify who owns AI risk decisions, who approves launch readiness, and what evidence is needed before executives can make a go/no-go decision.

The platform must also follow Design World's restrictions on AI usage. Unvetted models and shadow tools are not allowed. External LLM APIs may be allowed, but only within defined limits. Proprietary model fine-tuning is also not allowed, which means the platform should not be assumed to train proprietary models using Design World's internal data.

Policy requirements must also address acceptable use, data handling, user access, tool usage, and accountability. These requirements are important because the platform may interact with sensitive internal data and may support actions that go beyond basic items.

## 6. Data Classification and Handling Constraints

Design World should maintain a formal data classification framework. This framework should outline the information assets that are accessed or processed by AI. Data can span multiple sensitivity levels: public, internal use, confidential, and highly restricted.

Key Data Classification and Handling Constraints for the AI Platform:

- Existing Data Governance (SO-6):** The AI platform should adhere to previously established data governance, classification, and handling policies. Retrieval, storage, processing, and output of data should adhere to the rules of the category classification. Confidential and restricted data should remain under strict protection.
- Confidentiality and Privacy Protection (SO-1):** Strict access controls are required to protect undisclosed intellectual property. The AI platform must enforce role-based and context-aware access controls. No internal data should be transmitted to external machines, APIs, or AI models beyond approved cases.



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- **RAG Constraints:** Data should not be transmitted to users or systems with lower data clearance levels than the respective classification. A user or system should not be able to access data with higher data clearance levels via AI prompts or workarounds.
- **Agentic Workflow Limitations (SO-4):** Any actions taken by the AI must operate strictly within the user's permissions and data classification boundaries. The platform is prohibited from writing or modifying internal databases to prevent violations in data integrity or classification rules unless given explicit permission and is audited.
- **Storage and Retention:** Any caching or storage of data within the AI must follow Design World's data handling and retention policies. Persistent storage of sensitive data is prohibited.

These constraints protect the company by ensuring that the AI platform does not create new risks concerning data handling and classification. Design World's data governance must be present inside the AI workflow.

## 7. Compliance and Regulatory Considerations

A list of resources was gathered to help facilitate Design World's new data governance policies and compliance efforts. These resources provide a strong foundation of knowledge to design, update, protect, and maintain AI. Design World must also adhere to its previous compliance requirements and efforts.

- NIST AI Risk Management Framework (AI RMF)
- ISO/IEC 23894
- OWASP Top 10 for Large Language Model Applications
- MITRE ATLAS

Key Areas for AI Compliance Efforts:

- **Policy Alignment:** The AI platform must ensure that AI models are trustworthy and adhere to Design World's governance and compliance standards.
- **Auditing:** Logging, traceability, and explainability are features necessary to ensure that audits can be conducted to ensure compliance efforts are meaningful.
- **Intellectual Property Protection:** Controls must be implemented to ensure that exposure or transfer of sensitive information is prevented.

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- National Regulatory Requirements: Relevant data protection laws must be followed and enforced to the best of Design World's knowledge and ability.

## 8. Assumptions, Dependencies, and Open Questions

The following assumptions, dependencies, and open questions establish the baseline conditions for the remainder of the AI security engagement. These items are derived from the Design World case study, Project Change Request (PCR), and Statement of Work (SoW), and identify the organizational conditions, project dependencies, and areas requiring clarification before architecture, threat modeling, and assurance activities begin.

### a. Assumptions

**A-1:** Design World will deploy its enterprise AI platform within the approved Akamai/Linode environment using the infrastructure defined in the Statement of Work and Project Change Request.

**A-2:** The AI platform will consist of a conversational assistant supported by retrieval-augmented generation (RAG) and controlled agentic workflows to improve organizational productivity, knowledge access, and workflow automation.

**A-3:** Executive leadership will require documented evidence demonstrating that governance, security, and assurance objectives have been satisfied before authorizing production deployment through the Go/No-Go decision process.

**A-4:** The engagement will remain within the approved project scope. Changes to the cloud provider, expansion of system scope, or proprietary model fine-tuning are outside the scope of this engagement unless approved through a formal Project Change Request.

**A-5:** Design World will continue to apply its existing organizational data governance and classification policies throughout the AI platform lifecycle.

### b. Dependencies

**D-1:** Continued participation from executive leadership, business stakeholders, IT personnel, and the AI Security Architecture and Governance team throughout the engagement.

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**D-2:** Availability of approved enterprise data sources required to support the retrieval-augmented generation (RAG) capabilities of the AI platform.

**D-3:** Availability of approved AI services and supporting infrastructure operating within the architectural constraints established by Design World.

**D-4:** Completion and approval of subsequent project deliverables, including the AI architecture review, threat model, assurance planning, red-team evaluation, and executive Go/No-Go decision package.

**D-5:** Continued alignment with organizational governance policies and applicable AI security frameworks throughout the engagement.

**D-6:** Continued availability of timely stakeholder feedback throughout the engagement to review interim deliverables, validate assumptions, and

### **c. Open Questions**

**OQ-1:** Which business units, departments, and user roles will be authorized to access the AI platform during the initial deployment?

**OQ-2:** What categories of organizational data will be approved for retrieval, processing, and AI-generated responses?

**OQ-3:** What level of residual AI risk is considered acceptable by executive leadership before approving production deployment?

**OQ-4:** What audit logging, monitoring, and data retention requirements will apply to AI prompts, responses, and agentic workflow activities?

**OQ-5:** Are there additional contractual, customer, or international regulatory requirements that must be incorporated into the AI governance program?

**OQ-6:** What measurable business and operational success criteria will be used to evaluate the effectiveness of the AI platform following deployment?

**OQ-7:** How will Design World measure ongoing AI governance effectiveness after deployment, and what process will be used to periodically review and update governance requirements as business needs evolve?

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## 9. Conclusion

Design World's enterprise AI platform is intended to improve how employees access information, complete knowledge-based tasks, and support approved business workflows. However, the same capabilities that create business value also introduce security risk if the platform is not clearly defined before deployment. Because the system may retrieve organizational data and support executive level decisions, Design World must understand the platform's operating constraints before moving into architecture or assurance activities.

This report establishes the initial requirements baseline for the engagement. The business objectives, use cases, security objectives, governance expectations, data handling constraints, compliance considerations, assumptions, dependencies, and open questions provide a common foundation for the work that follows. By defining these requirements early, Design World is better positioned to evaluate whether the AI platform can support business goals while protecting sensitive information and maintaining accountability over AI-assisted activity.

## 10. Approvals

| Role              | Printed Name     | Signature               | Date     |
|-------------------|------------------|-------------------------|----------|
| Project Lead      | Shivam Patel     | <i>Shivam Patel</i>     | 06/29/26 |
| Security Analyst  | O.Kingsley Ero   | <i>O.Kingsley Ero</i>   | 06/29/26 |
| Security Analyst  | Mohammad Ahmed   | <i>Mohammad Ahmed</i>   | 06/29/26 |
| Security Engineer | Joshua Remington | <i>Joshua Remington</i> | 06/29/26 |
| Security Engineer | Brian Che        | <i>Brian Che</i>        | 06/29/26 |
| Network Engineer  | Mikailou Nikiema | <i>Mikailou Nikiema</i> | 06/29/26 |
| DW CEO            |                  |                         |          |